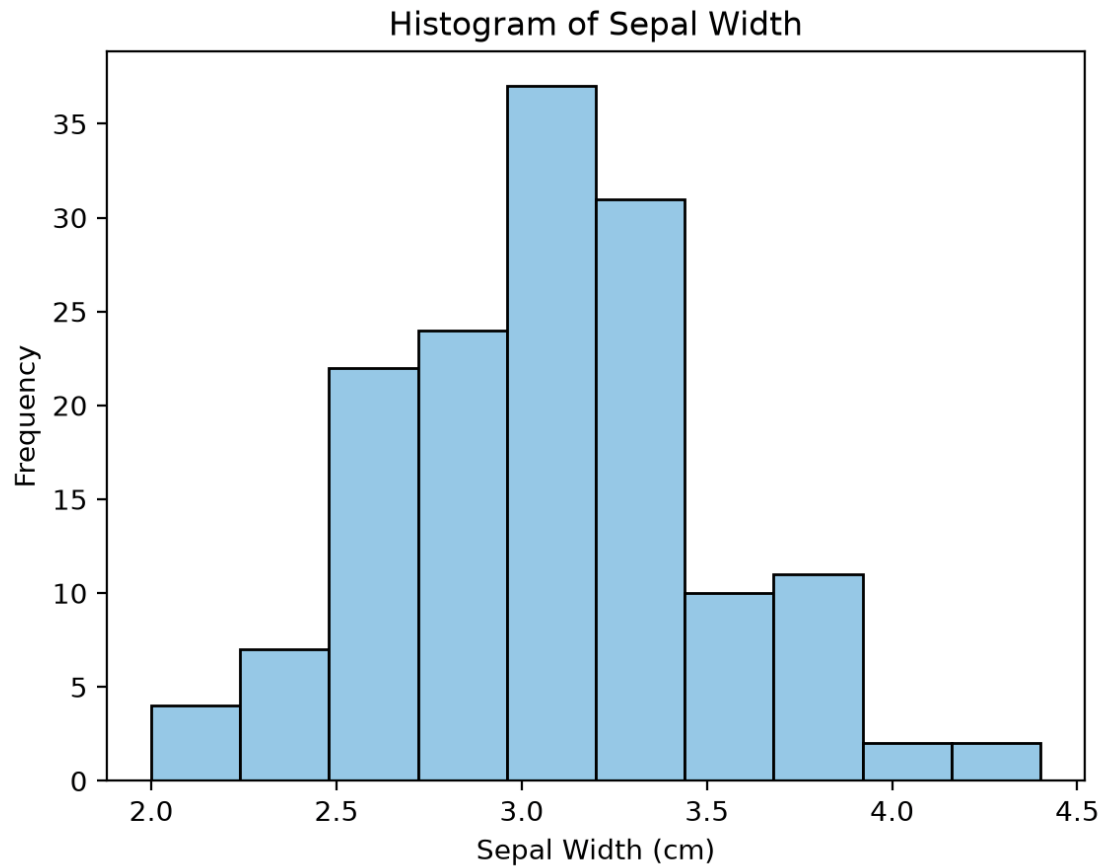


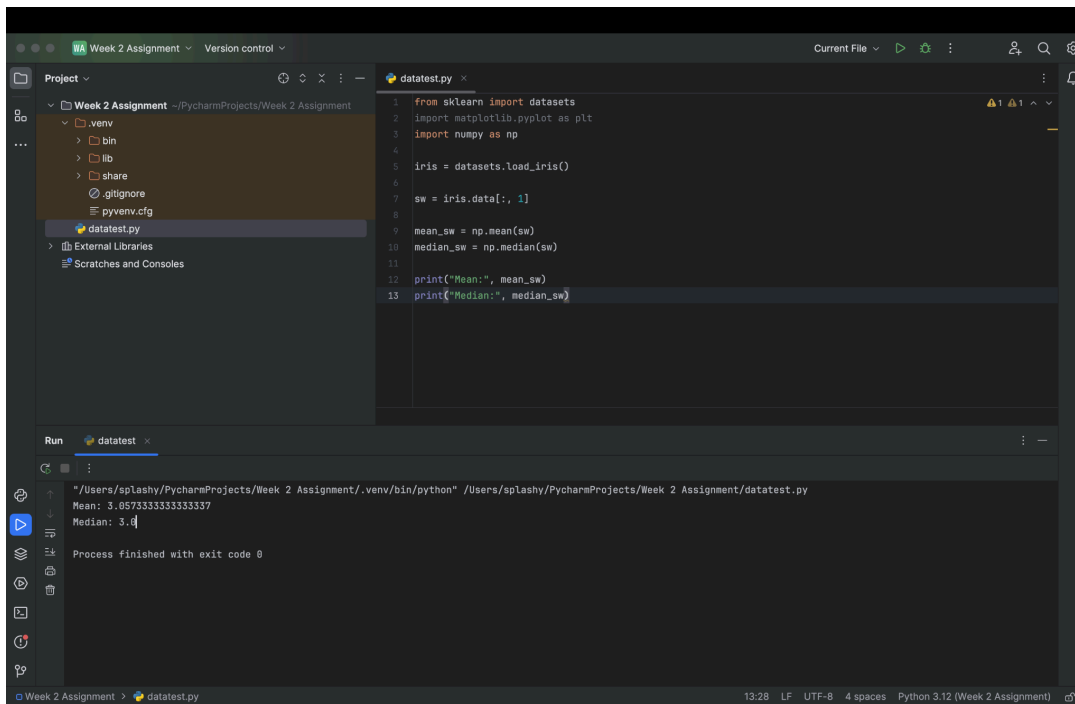
Week 3 Assignment 1



Based on the histogram from #1a, which would you expect to be higher, the mean or the median? Why?

Based on the histogram, I would expect the mean to be slightly higher than the median because it looks like there is a slightly higher frequency on the right side (right-skewed distribution).

Confirm your answer to #1b by actually finding these values.



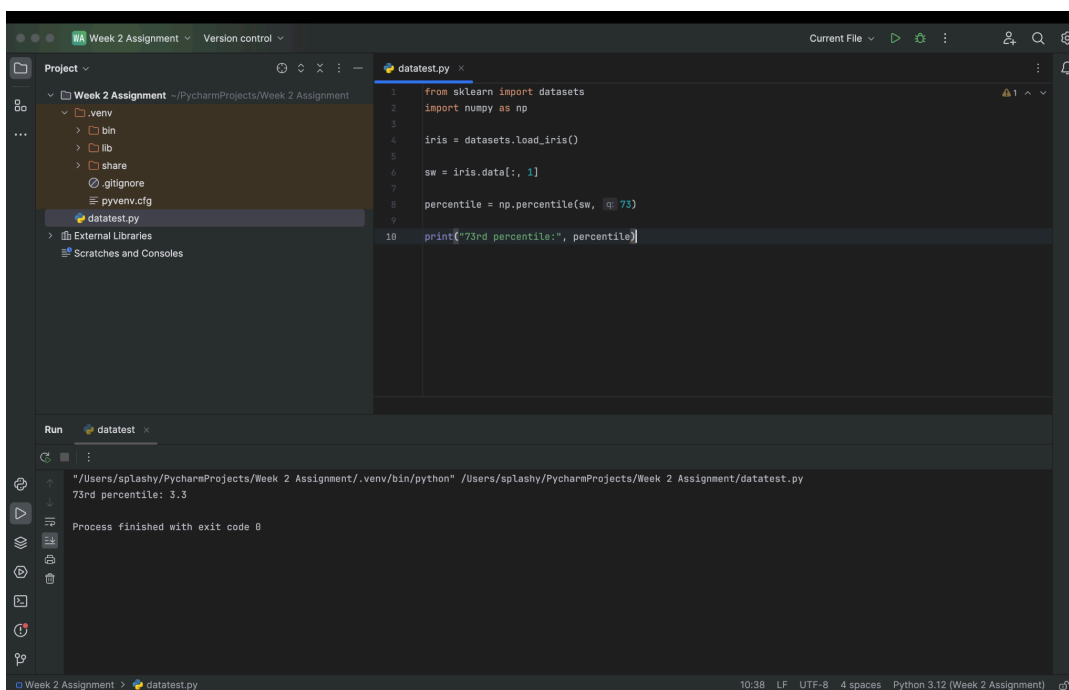
```
1 from sklearn import datasets
2 import matplotlib.pyplot as plt
3 import numpy as np
4
5 iris = datasets.load_iris()
6
7 sw = iris.data[:, 1]
8
9 mean_sw = np.mean(sw)
10 median_sw = np.median(sw)
11
12 print("Mean:", mean_sw)
13 print("Median:", median_sw)
```

Run dataatest

```
"/Users/splashy/PycharmProjects/Week 2 Assignment/.venv/bin/python" /Users/splashy/PycharmProjects/Week 2 Assignment/dataatest.py
Mean: 3.0573333333333337
Median: 3.4
Process finished with exit code 0
```

Only 27% of the flowers have a Sepal.Width higher than _____ cm.

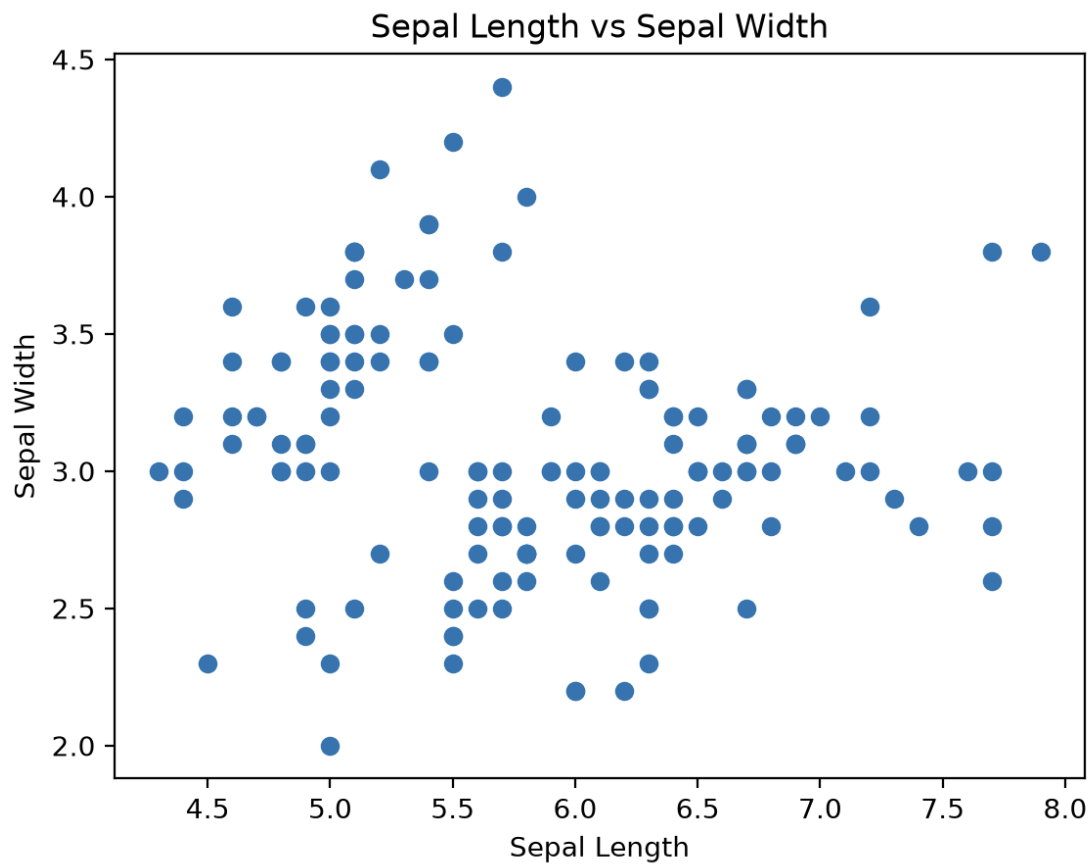
(3.3 cm)

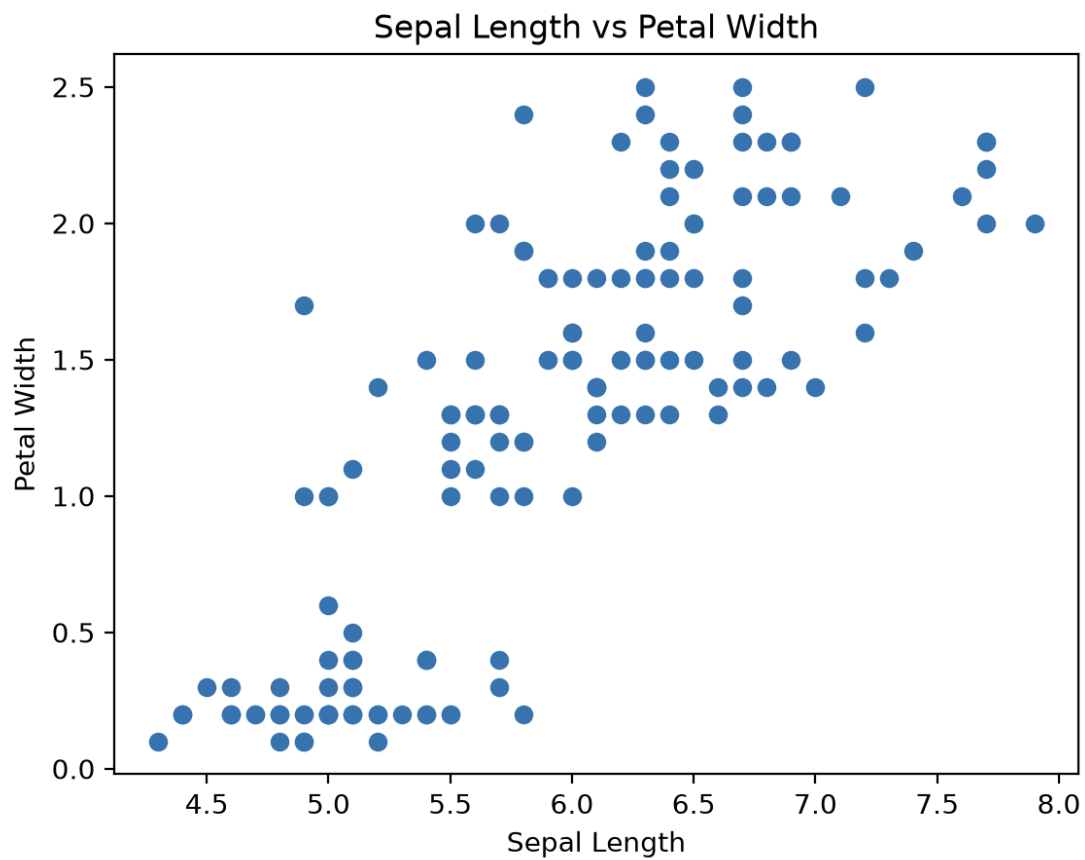


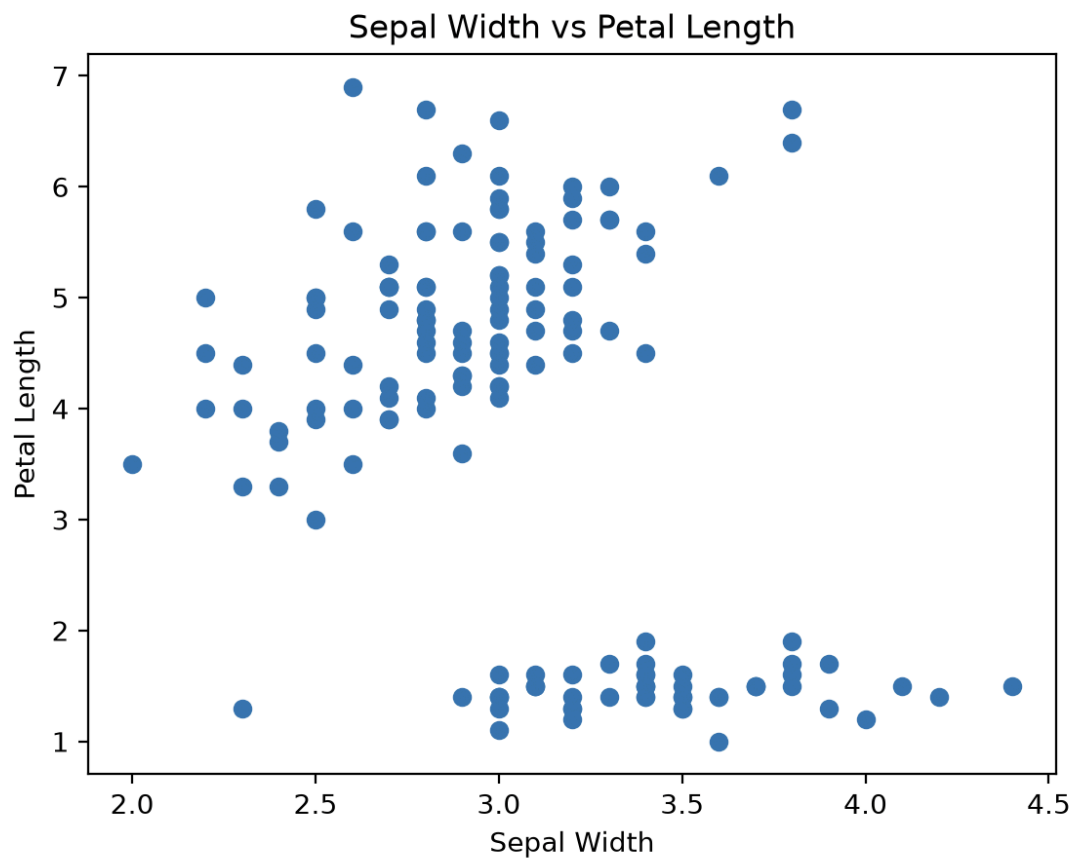
```
1 from sklearn import datasets
2 import numpy as np
3
4 iris = datasets.load_iris()
5
6 sw = iris.data[:, 1]
7
8 percentile = np.percentile(sw, 73)
9
10 print("73rd percentile:", percentile)
```

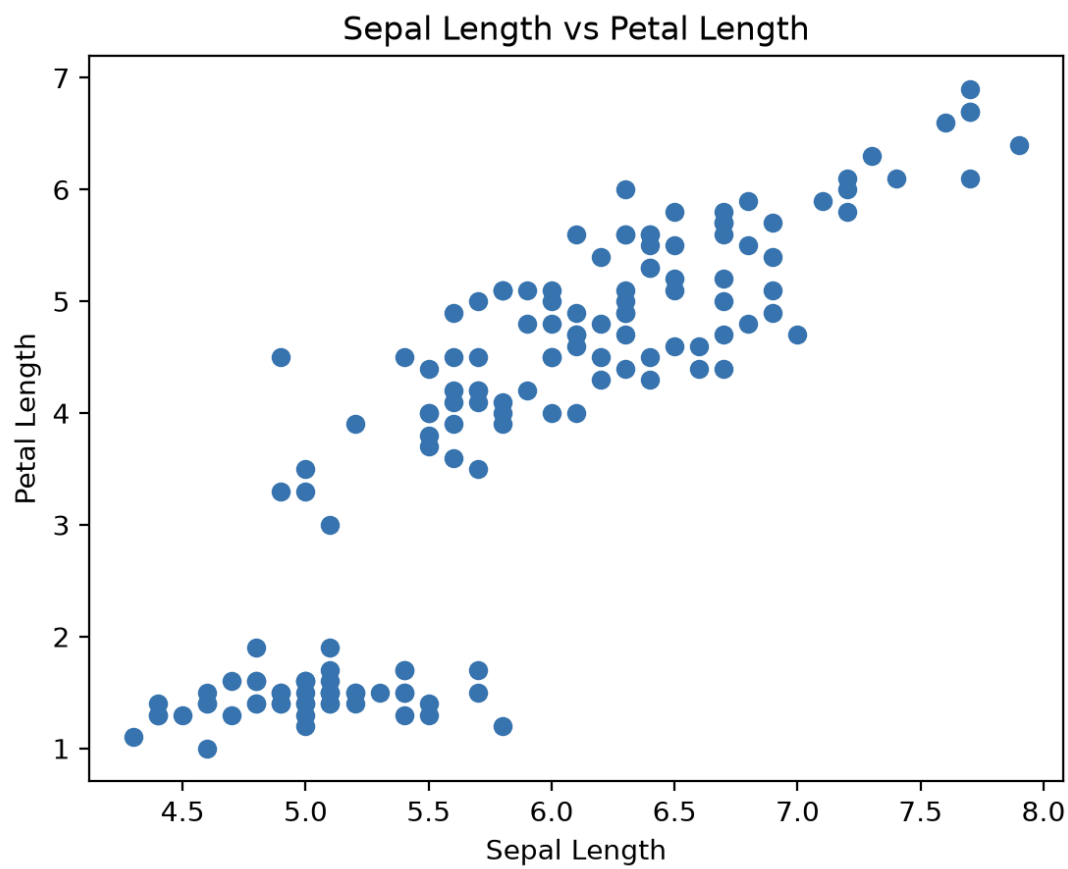
Run dataatest

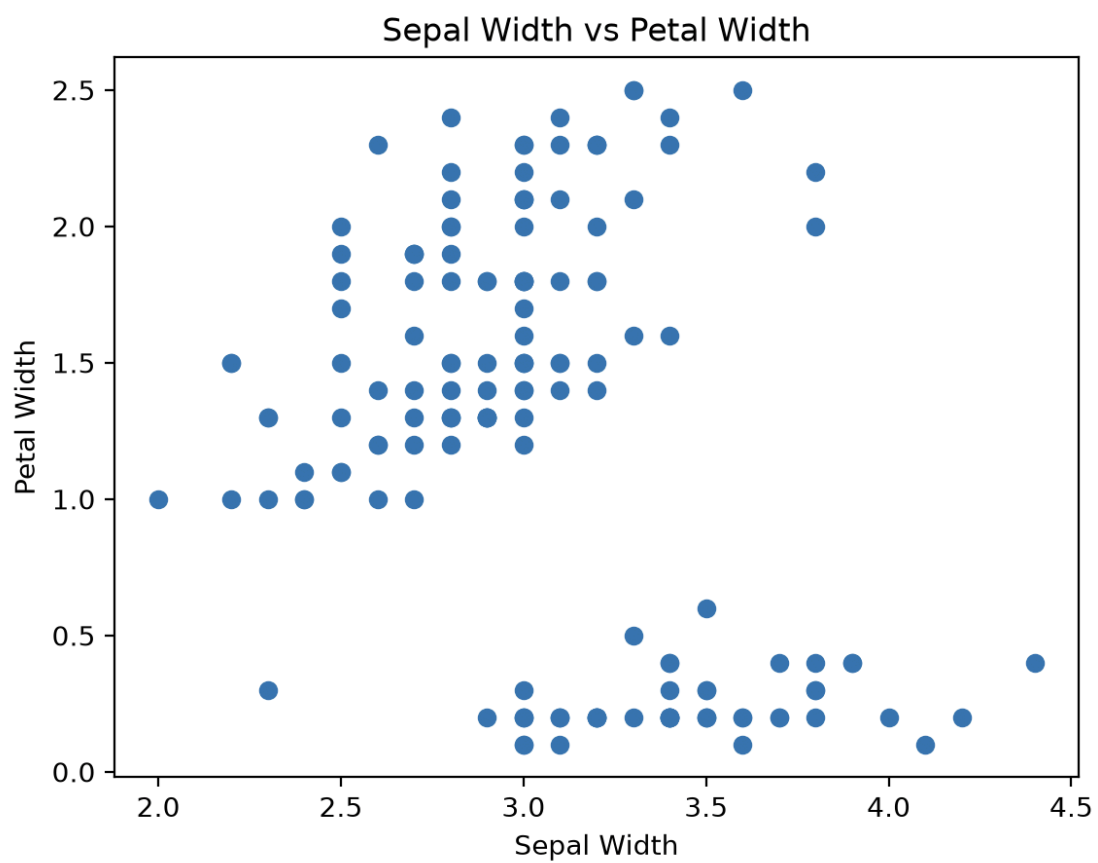
```
"/Users/splashy/PycharmProjects/Week 2 Assignment/.venv/bin/python" /Users/splashy/PycharmProjects/Week 2 Assignment/dataatest.py
73rd percentile: 3.3
Process finished with exit code 0
```

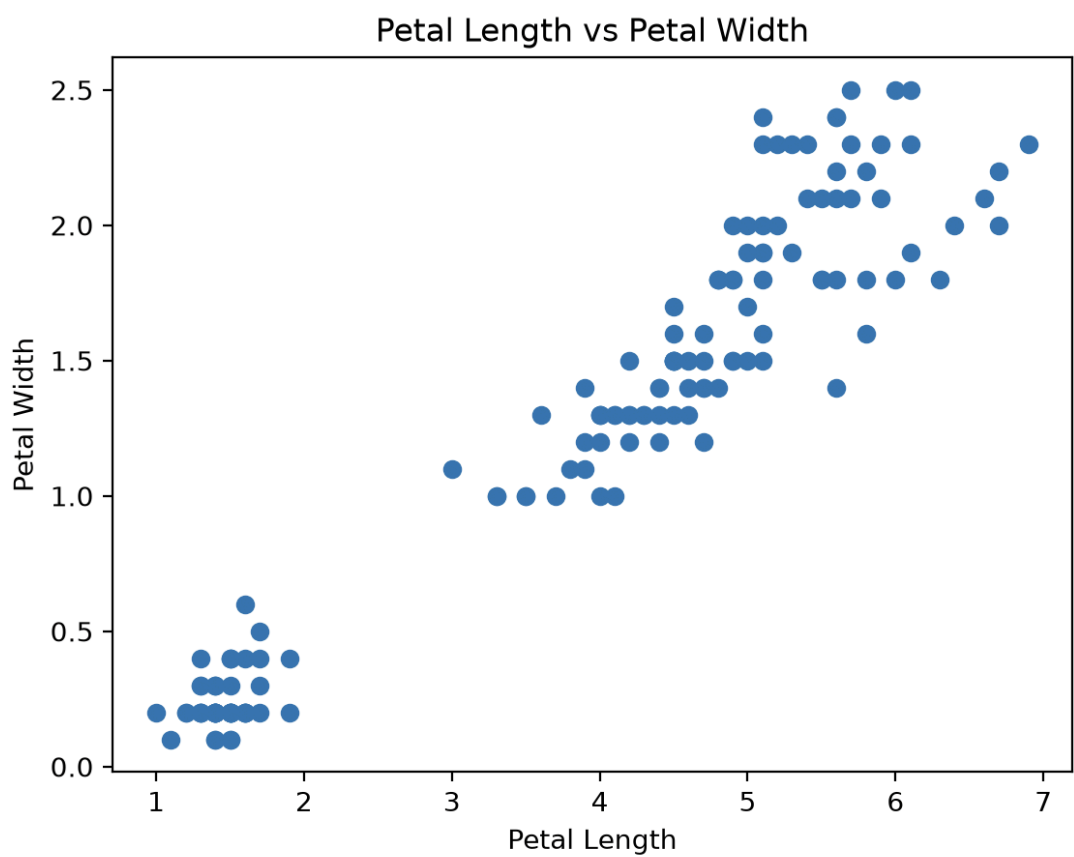








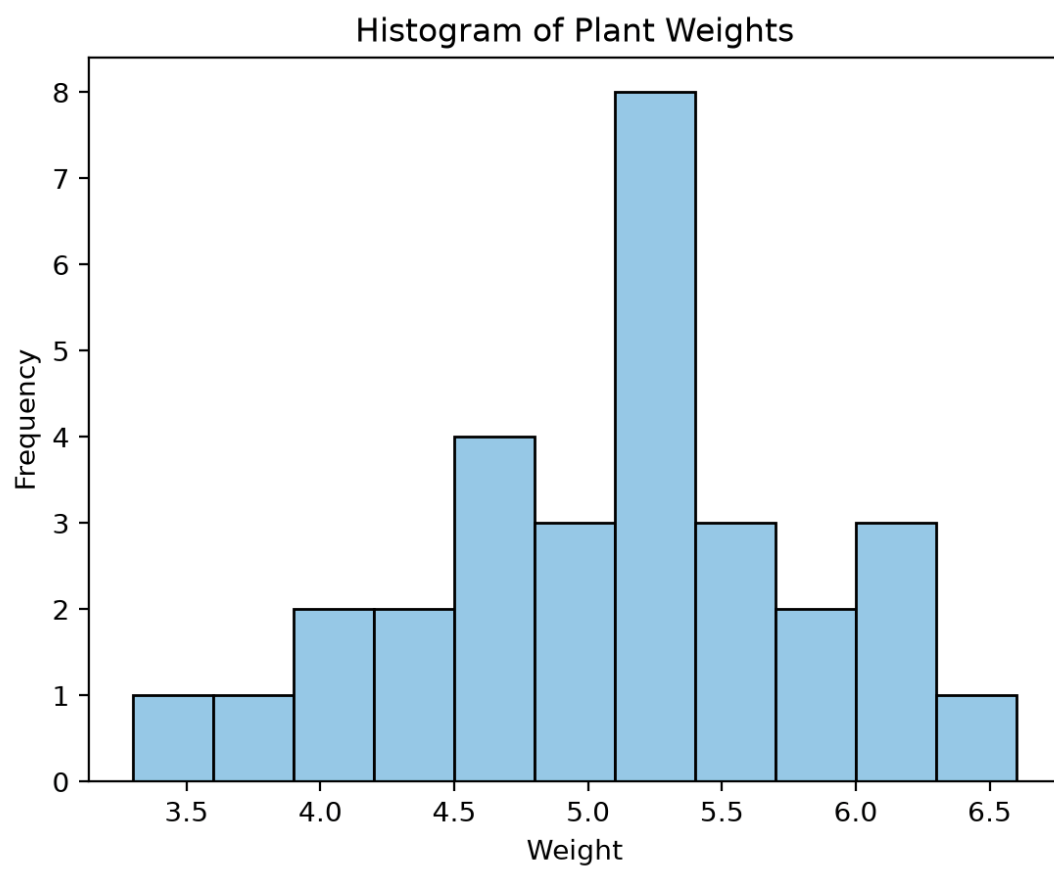


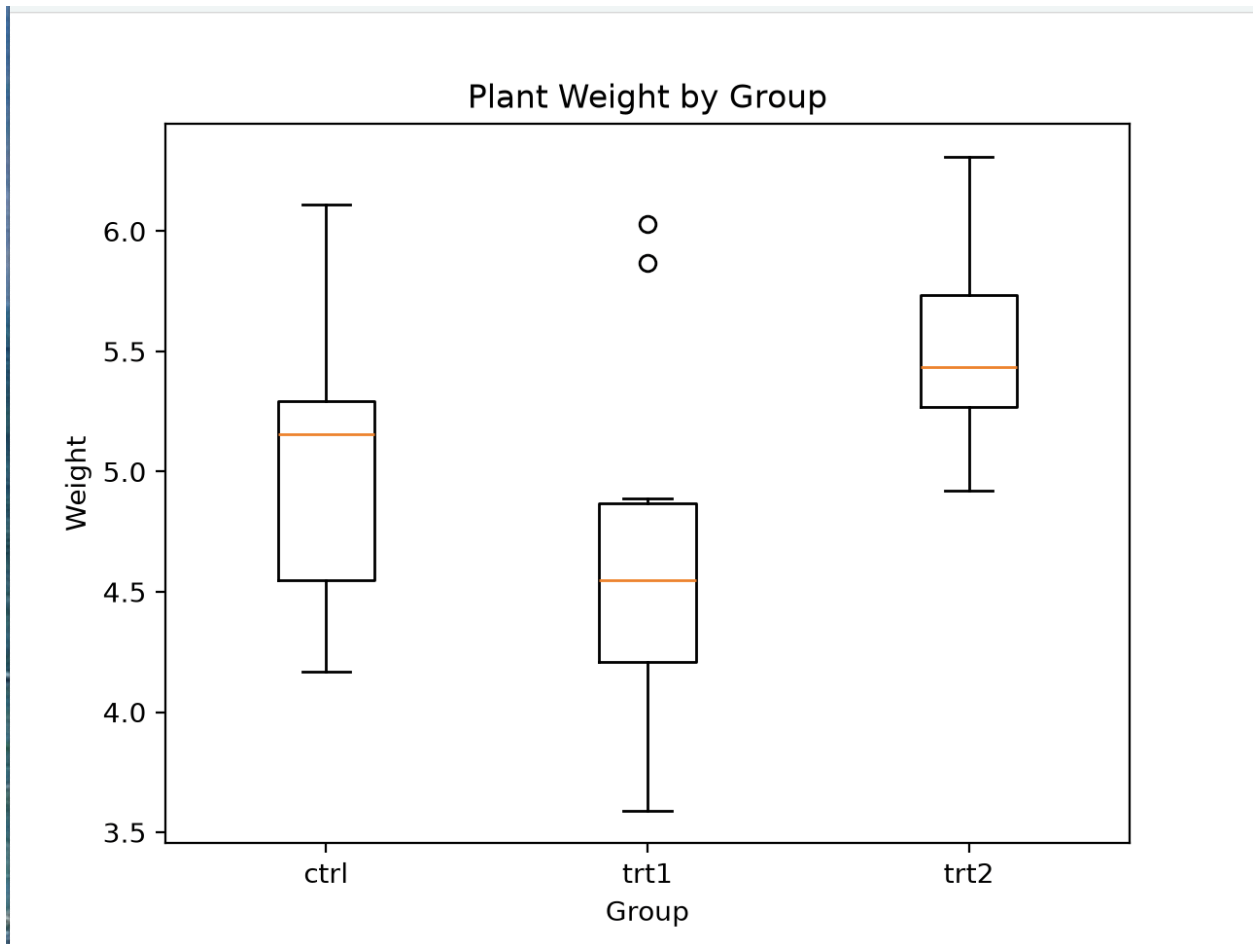



```
dataest.py x
12 # 1. Slengneth vs Swidth
13 plt.scatter(sl, sw)
14 plt.xlabel("Sepal Length")
15 plt.ylabel("Sepal Width")
16 plt.title("Sepal Length vs Sepal Width")
17 plt.show()
18
19 # 2. Slength vs Plength
20 plt.scatter(sl, pl)
21 plt.xlabel("Sepal Length")
22 plt.ylabel("Petal Length")
23 plt.title("Sepal Length vs Petal Length")
24 plt.show()
25
26 # 3. Slength vs Pwidth
27 plt.scatter(sl, pw)
28 plt.xlabel("Sepal Length")
29 plt.ylabel("Petal Width")
30 plt.title("Sepal Length vs Petal Width")
31 plt.show()
32
33 # 4. SWidth vs Plength
34 plt.scatter(sw, pl)
35 plt.xlabel("Sepal Width")
36 plt.ylabel("Petal Length")
37 plt.title("Sepal Width vs Petal Length")
38 plt.show()
39
40 # 5. SWidth vs Pwidth
41 plt.scatter(sw, pw)
42 plt.xlabel("Sepal Width")
43 plt.ylabel("Petal Width")
44 plt.title("Sepal Width vs Petal Width")
45 plt.show()
```

Based on #1e, which two variables appear to have the strongest relationship? And which two appear to have the weakest relationship?

Petal Length and Petal Width appear to have the strongest relationship because the points follow a clear upward pattern. Sepal Length and Sepal Width appear to have the weakest relationship because the points are more scattered and do not follow a clear pattern.



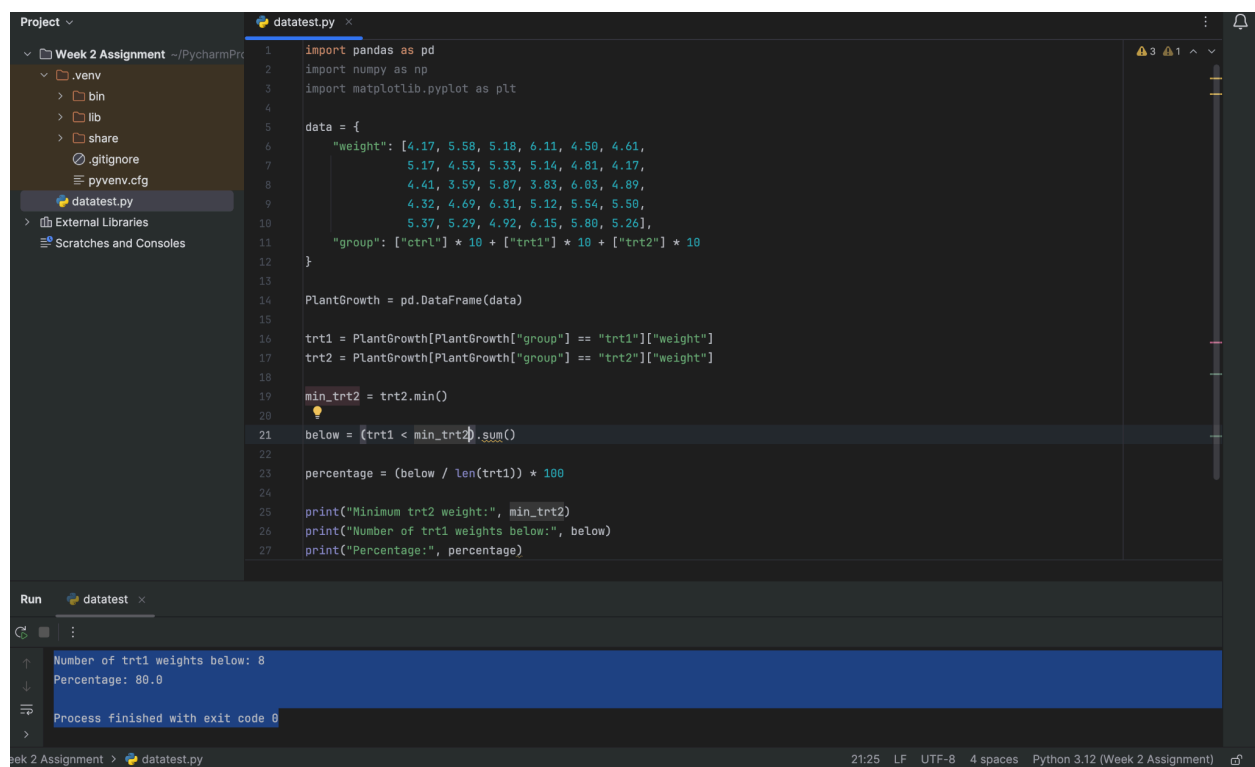


```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 data = {
6     "weight": [4.17, 5.58, 5.18, 6.11, 4.50, 4.61,
7               5.17, 4.53, 5.33, 5.14, 4.81, 4.17,
8               4.41, 3.59, 5.87, 3.83, 6.03, 4.89,
9               4.32, 4.69, 6.31, 5.12, 5.54, 5.50,
10              5.37, 5.29, 4.92, 6.15, 5.80, 5.26],
11     "group": ["ctrl"] * 10 + ["trt1"] * 10 + ["trt2"] * 10
12 }
13
14 PlantGrowth = pd.DataFrame(data)
15
16 ctrl = PlantGrowth[PlantGrowth["group"] == "ctrl"]["weight"]
17 trt1 = PlantGrowth[PlantGrowth["group"] == "trt1"]["weight"]
18 trt2 = PlantGrowth[PlantGrowth["group"] == "trt2"]["weight"]
19
20 plt.boxplot([ctrl, trt1, trt2],
21             tick_labels=["ctrl", "trt1", "trt2"])
22
23 plt.xlabel("Group")
24 plt.ylabel("Weight")
25 plt.title("Plant Weight by Group")
26
27 plt.show()
```

Based on the boxplots in #2b, approximately what percentage of the "trt1" weights are below the minimum "trt2" weight?

About 75% of the trt1 weights are below the minimum trt2 weight.

Find the exact percentage of the "trt1" weights that are below the minimum "trt2" weight.

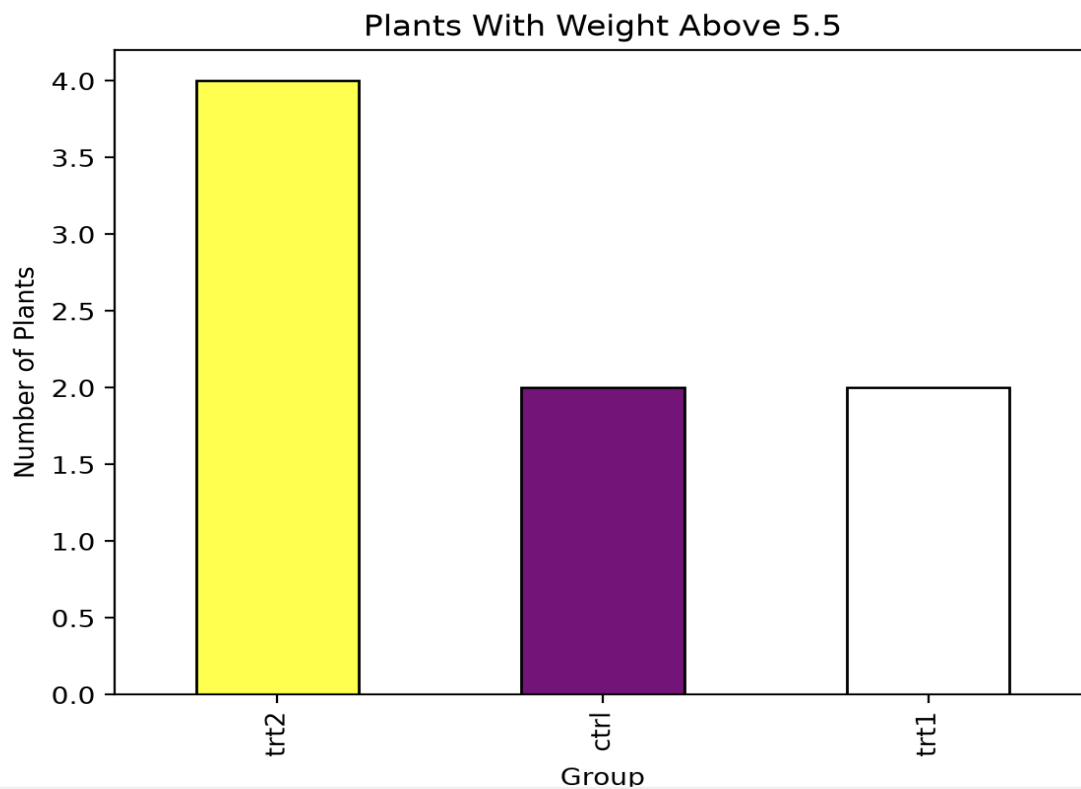


```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5 data = {
6     "weight": [4.17, 5.58, 5.18, 6.11, 4.50, 4.61,
7               5.17, 4.53, 5.33, 5.14, 4.81, 4.17,
8               4.41, 3.59, 5.87, 3.83, 6.03, 4.89,
9               4.32, 4.69, 6.31, 5.12, 5.54, 5.50,
10              5.37, 5.29, 4.92, 6.15, 5.80, 5.26],
11     "group": ["ctrl"] * 10 + ["trt1"] * 10 + ["trt2"] * 10
12 }
13
14 PlantGrowth = pd.DataFrame(data)
15
16 trt1 = PlantGrowth[PlantGrowth["group"] == "trt1"]["weight"]
17 trt2 = PlantGrowth[PlantGrowth["group"] == "trt2"]["weight"]
18
19 min_trt2 = trt2.min()
20
21 below = (trt1 < min_trt2).sum()
22
23 percentage = (below / len(trt1)) * 100
24
25 print("Minimum trt2 weight:", min_trt2)
26 print("Number of trt1 weights below:", below)
27 print("Percentage:", percentage)
```

Run datatest

```
Number of trt1 weights below: 8
Percentage: 80.0
Process finished with exit code 0
```

Week 2 Assignment > datatest.py 21:25 LF UTF-8 4 spaces Python 3.12 (Week 2 Assignment)



```
data = {
    "weight": [4.17, 5.58, 5.18, 6.11, 4.50, 4.61,
               5.17, 4.53, 5.33, 5.14, 4.81, 4.17,
               4.41, 3.59, 5.87, 3.83, 6.03, 4.89,
               4.32, 4.69, 6.31, 5.12, 5.54, 5.50,
               5.37, 5.29, 4.92, 6.15, 5.80, 5.26],
    "group": ["ctrl"] * 10 + ["trt1"] * 10 + ["trt2"] * 10
}

PlantGrowth = pd.DataFrame(data)

above_55 = PlantGrowth[PlantGrowth["weight"] > 5.5]

group_counts = above_55["group"].value_counts()

group_counts.plot(
    kind="bar",
    color=["yellow", "purple", "white"], #vikings_colors!
    edgecolor="black"
)

plt.xlabel("Group")
plt.ylabel("Number of Plants")
plt.title("Plants With Weight Above 5.5")
plt.show()
```